

Predicting ICU Mortality with MIMIC-III

A reproducible Python workflow for cohort design, leakage control, baseline ML, and model interpretation.

by: JC Diamante

Cohort

49,695 adult ICU admissions

Outcome

5,739 deaths (11.5%)

Code link

<https://github.com/Zeraphim/mimic-iii-predicting-adult-icu-mortality>

What can we predict in the first ICU day?

Task: predict in-hospital mortality for adult ICU admissions.

Prediction time: first 24 hours after ICU admission.

Why this matters: it teaches cohort construction, time windows, missingness, class imbalance, and interpretability.

Why not diagnoses: ICD codes are usually assigned after the stay and can leak outcome information.

Label

ADMISSIONS.HOSPITAL_EXPIRE_FLAG

Primary unit

First ICU stay per hospital admission

Leakage rule

Use only data available by hour 24

From protected CSVs to a modeling table

1. Cohort

ICUSTAYS + ADMISSIONS
+ PATIENTS

2. Window

Labs within 0-24h after
ICU intime

3. Features

Demographics, admission
context, lab summaries

4. Models

Logistic regression
+ random forest

Adult cohort excludes NICU and patients **under 18**.

Ages over 89 are **capped at 90** because MIMIC shifts dates for older patients.

First-day labs are summarized with **min, max, mean**, and **count**.

The feature cache avoids rescanning the large **LABEVENTS** file on every rerun.

Adult ICU mortality cohort

Rows

49,695 first ICU admissions

Deaths

5,739 in-hospital deaths

Rate

11.5% mortality

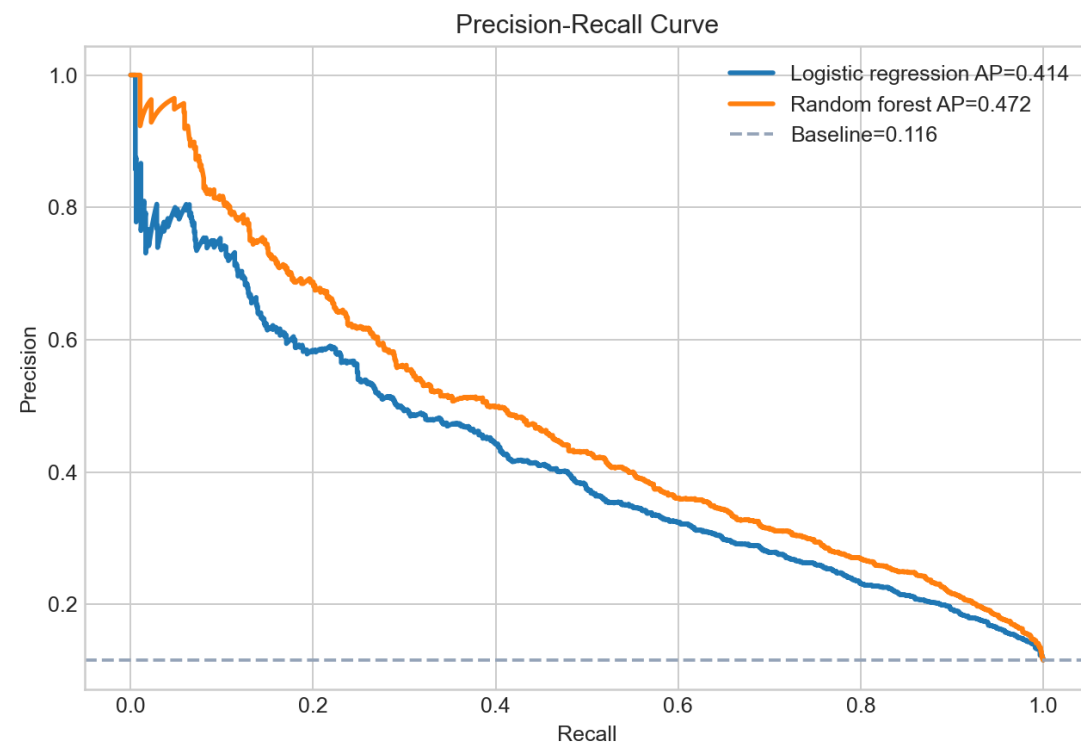
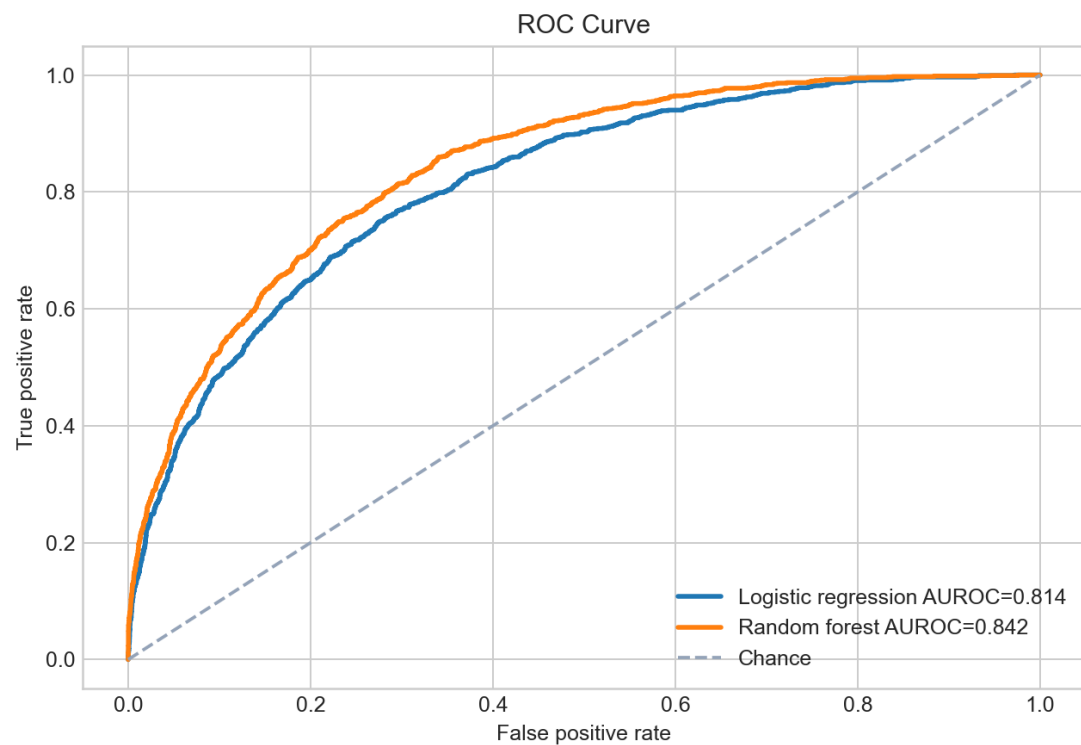
The outcome is **imbalanced**, so accuracy is not enough.

AUROC describes ranking across thresholds.

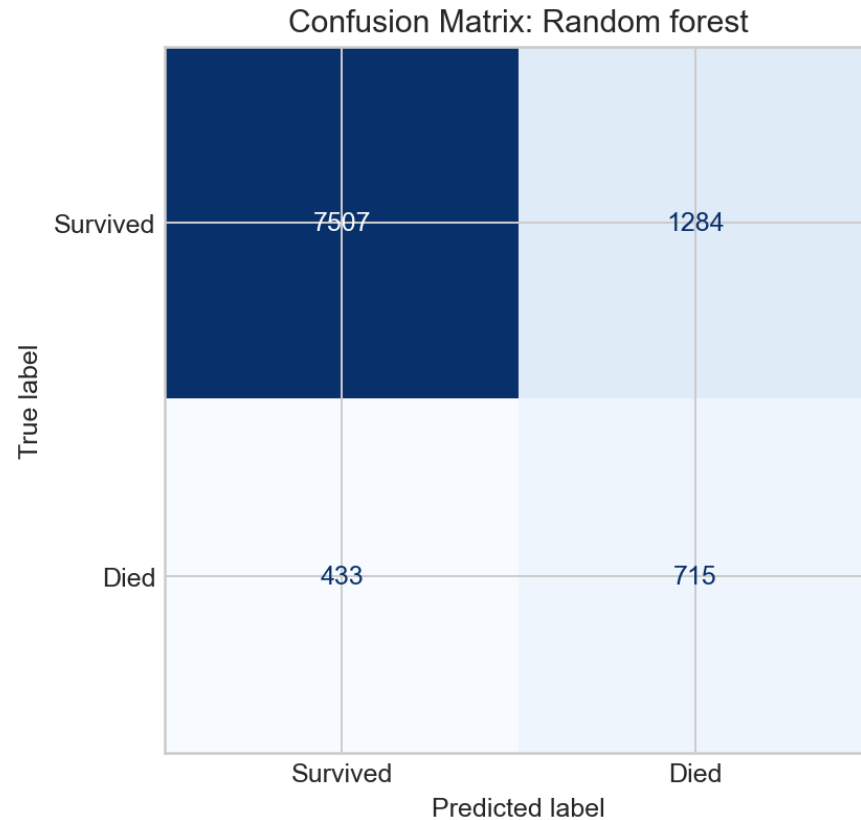
Average precision is **more sensitive** to performance on the death class.

Balanced accuracy keeps both classes visible at the **default threshold**.

Model discrimination



Default-threshold behavior



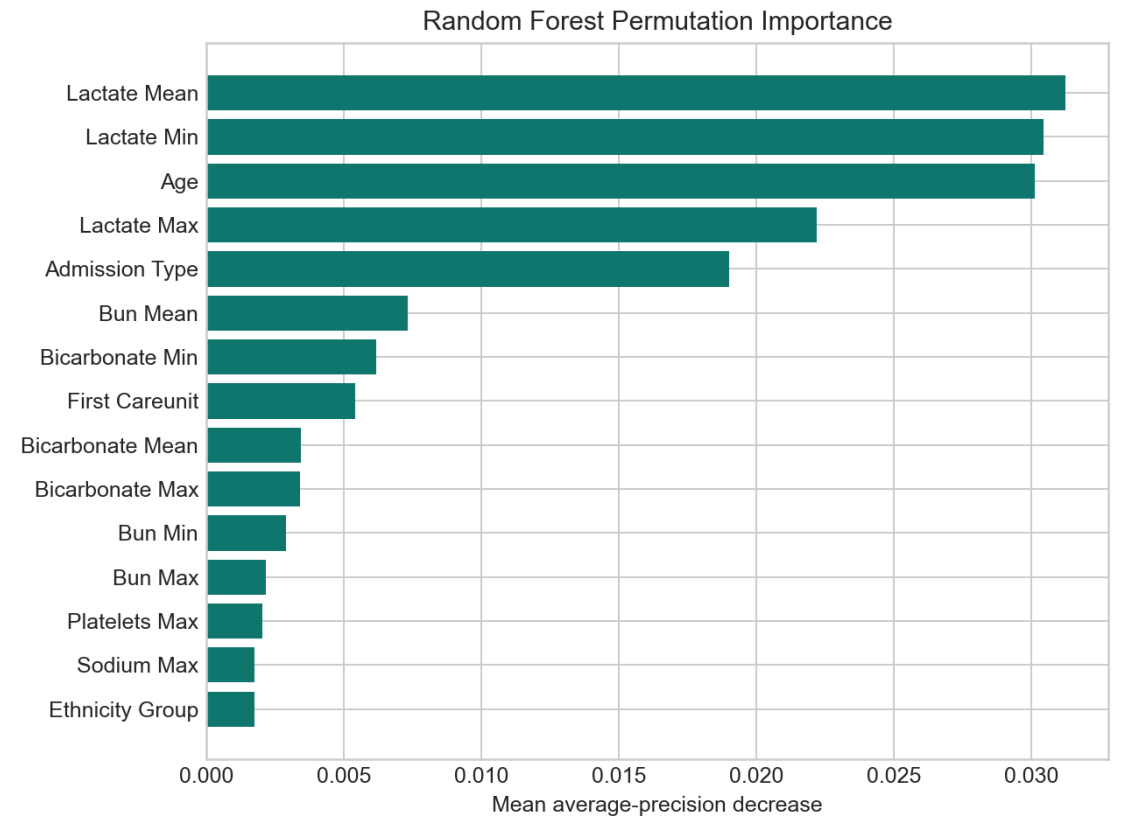
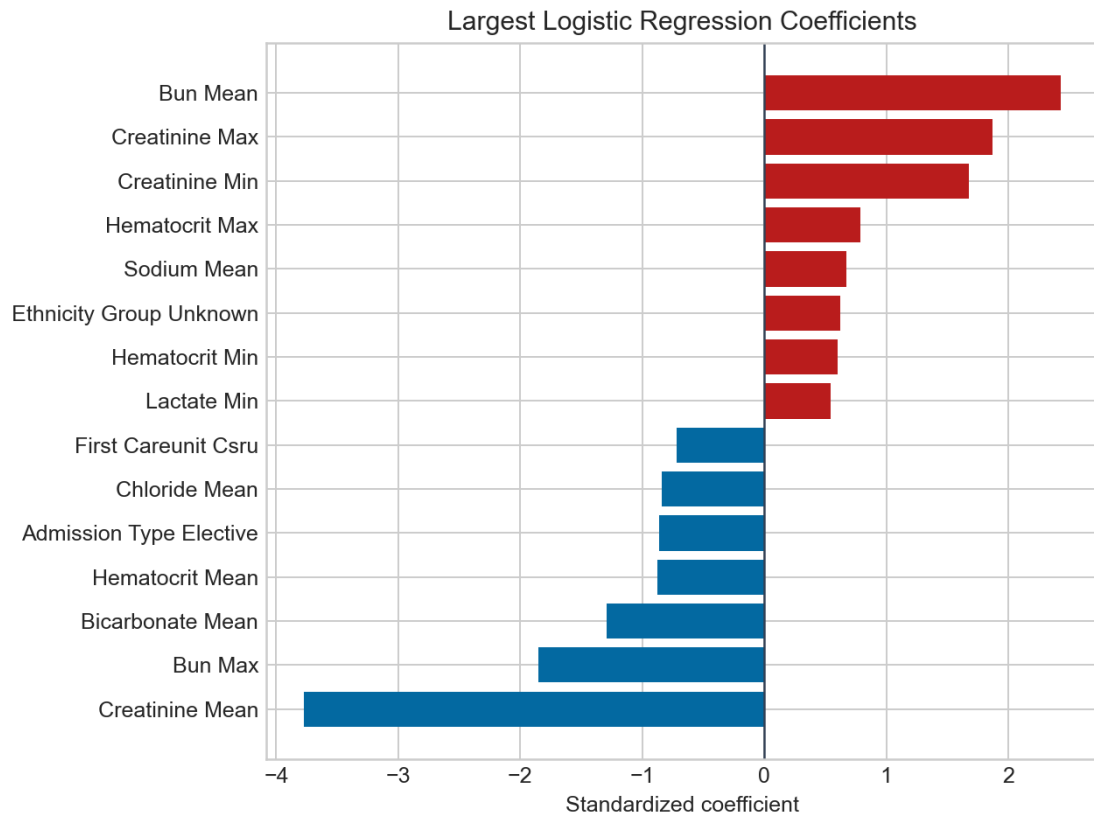
Best model by average precision: Random forest.

AUROC: 0.842.

Average precision: 0.472.

A clinical workflow would tune the threshold for the cost of false positives vs. false negatives.

What features drive the baselines?



How a peer can reproduce the tutorial

1. Activate the course Python environment: conda activate masters.
2. Keep MIMIC-III CSV files in Datasets/mimiciii/1.4.
3. Run mimic_mortality_tutorial.py to generate the feature cache, metrics, and figures.
4. Open MIMIC_III_Mortality_ML_Tutorial.ipynb for the step-by-step tutorial.
5. Do not publish raw MIMIC data or row-level patient excerpts.

Submission files

Slides PDF + GitHub Repository

Anonymity

Remove names, personal paths, and identifying repo metadata